

2 Image Representation

2.1 Introduction

This chapter centers around the question of how to represent the information contained in images. Together with the next two chapters it lays the mathematical foundations for low-level image processing. Two key points are emphasized in this chapter.

First, the information contained in images can be represented in entirely different ways. The most important are the spatial representation (Section 2.2) and wave number representation (Section 2.3). These representations just look at spatial data from different points of view. Since the various representations are complete and equivalent, they can be converted into each other. The conversion between the spatial and wave number representation is the well-known *Fourier transform*. This transform is an example of a more general class of operations, the *unitary transforms* (Section 2.4).

Second, we discuss how these representations can be handled with digital computers. How are images represented by arrays of digital numbers in an adequate way? How are these data handled efficiently? Can fast algorithms be devised to convert one representation into another? A key example is the fast Fourier transform, discussed in Section 2.5.

2.2 Spatial Representation of Digital Images

2.2.1 Pixel and Voxel

Images constitute a spatial distribution of the *irradiance* at a plane. Mathematically speaking, the spatial irradiance distribution can be described as a continuous function of two spatial variables:

$$E(x_1, x_2) = E(\mathbf{x}). \quad (2.1)$$

Computers cannot handle continuous images but only arrays of digital numbers. Thus it is required to represent images as two-dimensional arrays of points. A point on the 2-D grid is called a *pixel* or *pel*. Both words are abbreviations of the word picture element. A pixel represents the irradiance at the corresponding grid position. In the simplest case, the pixels are located on a rectangular grid. The position of the pixel

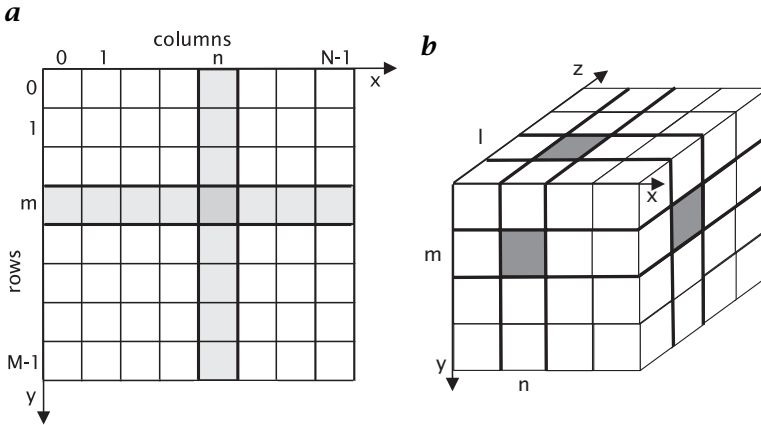


Figure 2.1: Representation of digital images by arrays of discrete points on a rectangular grid: **a** 2-D image, **b** 3-D image.

is given in the common notation for matrices. The first index, m , denotes the position of the row, the second, n , the position of the column (Fig. 2.1a). If the digital image contains $M \times N$ pixels, i.e., is represented by an $M \times N$ matrix, the index n runs from 0 to $N - 1$, and the index m from 0 to $M - 1$. M gives the number of rows, N the number of columns. In accordance with the matrix notation, the vertical axis (y axis) runs from top to bottom and not vice versa as it is common in graphs. The horizontal axis (x axis) runs as usual from left to right.

Each pixel represents not just a point in the image but rather a rectangular region, the elementary cell of the grid. The value associated with the pixel must represent the average irradiance in the corresponding cell in an appropriate way. Figure 2.2 shows one and the same image represented with a different number of pixels as indicated in the legend. With large pixel sizes (Fig. 2.2a, b), not only is the spatial resolution poor, but the gray value discontinuities at pixel edges appear as disturbing artifacts distracting us from the content of the image. As the pixels become smaller, the effect becomes less pronounced up to the point where we get the impression of a spatially continuous image. This happens when the pixels become smaller than the spatial resolution of our visual system. You can convince yourself of this relation by observing Fig. 2.2 from different distances.

How many pixels are sufficient? There is no general answer to this question. For visual observation of a digital image, the pixel size should be smaller than the spatial resolution of the visual system from a nominal observer distance. For a given task the pixel size should be smaller than the finest scales of the objects that we want to study. We generally find, however, that it is the available sensor technology (see Section 1.7.1)